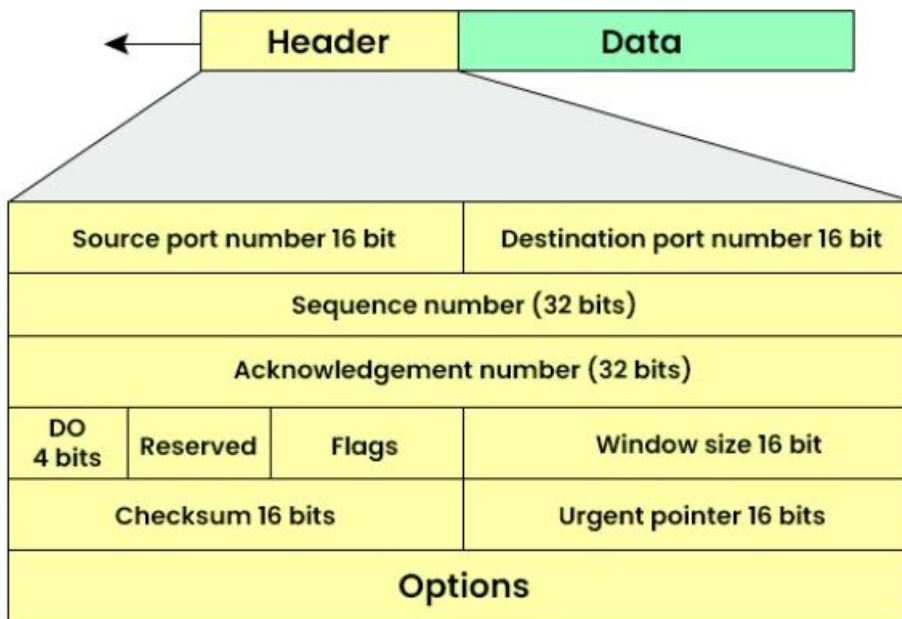


TCP Header Format



TCP Header Field

Function (1–2 Lines)

Source Port Number (16 bits)	Identifies the application that is sending the data. For example, a message sent from the WhatsApp app.
Destination Port Number (16 bits)	Identifies the application that should receive the data, such as the Instagram or WhatsApp server.
Sequence Number (32 bits)	Keeps the packets in the correct order so the receiver can rebuild the original message correctly.
Acknowledgement Number (32 bits)	Confirms that the previous packet has been received successfully and tells the sender what packet is expected next.
Data Offset (DO – 4 bits)	Shows the size of the TCP header and tells where the actual data begins.
Reserved	Reserved for future use. It is normally set to 0 .
Flags	Controls the TCP connection. Examples: SYN starts a connection, ACK confirms data, and FIN closes the connection.

TCP Header Field	Function (1–2 Lines)
Window Size (16 bits)	Tells the sender how much data the receiver can accept before sending another acknowledgement.
Checksum (16 bits)	Checks whether the data was damaged during transmission. If an error is found, the packet is sent again.
Urgent Pointer (16 bits)	Points to urgent data that should be processed first. It is used only when the URG flag is set.
Options	Stores extra information, such as Maximum Segment Size (MSS) and Window Scaling, to improve TCP performance.